

Object Oriented Programming

Introduction to Object Oriented Programming

Lesson 01

- What is **Programming**?
 - The process of writing computer programs.

- **What is Object Oriented Programming**
 - Object-oriented programming
 - is a paradigm
 - based on the concept of wrapping pieces of data, and behavior related to that data,
 - into special bundles called objects,
 - which are constructed from a set of “blueprints”,
 - defined by a programmer,
 - called classes

- Why Object Orientation ?
 - For keeping large software projects manageable by human programmers.
 - **Modularity**
 - **Information-hiding**

- **Code re-use**
- **Pluggability and debugging ease**
- **Modularity**
 - The source code for an object can be written and maintained independently of the source code for other objects.
 - Once created, an object can be easily passed around inside the system.
 -



Non-modular



Modular

- **Information-hiding**
 - By interacting only with an object's methods, the details of its internal implementation remain hidden from the outside world.



○ **Code re-use**

- If an object already exists (perhaps written by another software developer), you can use that object in your program.

(වස්තුවක් දැනටමත් පවතී නම් (සමහර විට වෙනත් මෘදුකාංග සංවර්ධකයෙකු විසින් ලියන ලද), ඔබට එම වස්තුව ඔබේ වඩාසටහනේ භවිතකළ හැක.)

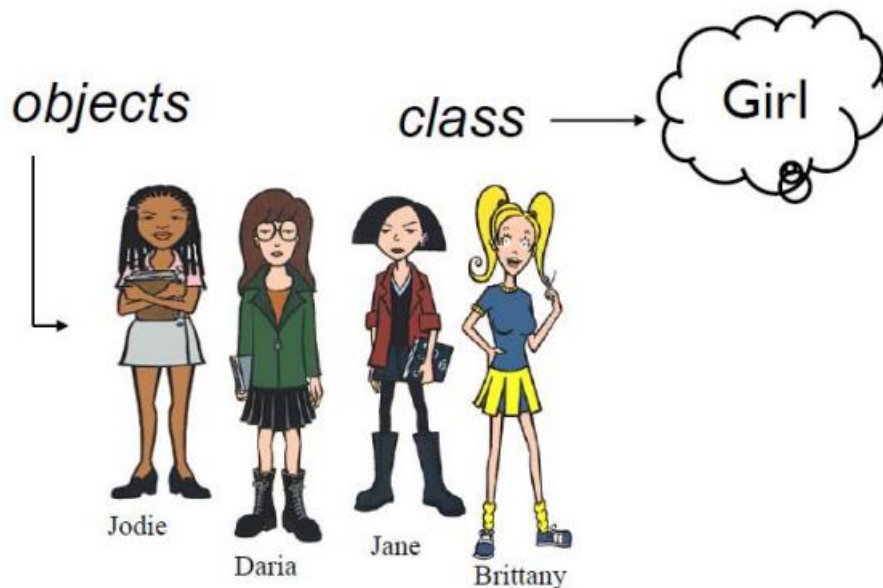
○ **Pluggability and debugging ease**

- If a particular object turns out to be problematic, you can simply remove it from your application and plug in a different object as its replacement.

(යම් වස්තුවක් ගැළපී සහගත බවට පත් වුවහොත්, ඔබට එය ඔබගේ යෙදුමෙන් ඉවත් කර එහි ප්‍රතිස්ථාපනය ලෙස වෙනත් වස්තුවක් සම්බන්ධ කළ හැක.)

➤ Objects and Classes

- Classes reflect(පිළිබිඹු කරයි) concepts, objects reflect instances that embody those concepts.



➤ Objects as instances of Classes

➤ Instantiation

- An Object is instantiated from a Class.

```
BankAccount myAccount;  
myAccount = new BankAccount();
```

```
BankAccount gatesAccount;  
gatesAccount = new BankAccount();
```

➤ **What Is an Object?**

- Object is an instance of a class.

➤ **Characteristics of Objects**

- State
- Behavior
- Identity

○ **Alan Kay's 5 rules**

- Everything is an object.
- A program is a bunch of objects telling each other what to do by sending messages.
- Each object has its own memory made up of other objects.
- Every object has a type.
- All objects of a particular type can receive the same messages.

(invoke methods)

➤ **What is a Class...**

- A blueprint of an object.

A class is the blueprint from which individual objects are created.

➤ Instance

- An instance is a specific realization of any object.

(උදාහරණයක් යනු ඕනෑම වස්තුවක් නිශ්චිතව අවබෝධ කර ගනීමකි.)

➤ Instantiation

- Instantiation is the creation of a real instance or particular realization of an abstraction or template, such as a class of objects or a computer process.
- (Instantiation යනු වස්තු පන්තියක් හෝ පරිගණක ක්‍රියාලියක් වැනි විශුක්ත හැඳුනුම්ලත් සඳහා අවස්ථාවක් හොඳින් ශේෂිත අවබෝධයක් නිර්මාණය කිරීමයි.)

➤ Advantages of OOP

- Ease of modeling real world in software context.(මුද්‍රාකරණ සන්දර්භය තුළ සම්ප්‍රදායික ආකෘති නිර්මාණය කිරීමේ පහසුව)

- Object-oriented systems can be easily upgraded from small to large scale.(වස්තු-නමුරු පද්ධති කුඩා පරිමාණයේ සිට විශාල පරිමාණය දක්වා පහසුවෙන් වඩිදියුණු කළ හැක.)
- It is easy to partition the work in a project based on objects.(වස්තු මත පදනම්ව ව්‍යාප්තියක වඩ කෙටස් කිරීම පහසුය)
- Object-oriented programming offers a new and powerful model for writing computer software.(වස්තු-නමුරු වඩසටහන්කරණය පරිගණක මෘදුකාංග ලිවීම සඳහානව සහ බලවත් ආකෘතියක් ඉදිරිපත් කරයි.)
- It reduces software maintenance and developing costs.(එය මෘදුකාංග නඩත්තු කිරීම සහ සංවර්ධනය කිරීමේ පිරිවැය අඩු කරයි.)

Summary

- What is Object Oriented Programming
- Fundamentals of Object Orientation
- Why Object Orientation ?
 - Modularity
 - Information-hiding

- Code re-use
- Pluggability and debugging ease
- Understanding Classes and Objects
- Real-World Scenario
- Class
- Object
- Instance
- Instantiation